

5 (a) when a wave passes through a slit / by an edge
the wave spreads out / changes direction M1
A1 [2]

(b) diagram: wavelength unchanged M1
wavefront flat at centre, curving into geometrical shadow A1 [2]

(c) $d \sin \theta = n\lambda$ C1
for $\theta = 90^\circ$
 $1 / (650 \times 10^3) = n \times 590 \times 10^{-9}$ M1
 $n = 2.6$
number of orders is 2 A1 [3]

(d) intensity / brightness decreases (as order increases) B1 [1]

6 (a) (i) either $P = V^2 / R$ or $P = VI$ and $V = IR$ C1